

Practicum Sprint #2 - Project Planning

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1 PROJECT TOPIC

1.1 Background

We'd like to focus on the medication safety of patients, especially those who take multiple prescriptions or over-the-counter drugs at the same time. Many medications may interact with one another, which increases the risk of side effects, reduces effectiveness, or can have a serious effect. Patients with allergies may unknowingly take medications with related ingredients that could trigger harmful reactions. Timing of medications is also important since some medications should not be taken together, while others need to be taken with food, without food, or staggered. It can be difficult for patients to recognize these safety issues because medication instructions tend to be long, complex, and difficult to find.

Our project focuses on creating a medication safety assistant that will help patients understand these risks more easily accessible. The application will allow users to input their current medications, allergy information, and medication schedules. It will analyze the information given for possible drug interactions, allergy conflicts, and timing conflicts. The application is designed to provide understandable warnings and explanations that will help users become more informed about what they're taking rather than diagnosing or prescribing medications. This application can help have an informed conversation between patients, pharmacists, and healthcare providers.

1.2 Justification

One study from 2014 found that an estimated 4 out of 1000 emergency department visits were for adverse drug events (Shehab N et. al). The most commonly implicated drugs were antibiotics, antipsychotics, anti-coagulants, diabetes agents, and opioid analgesics. These

adverse drug events can be life-threatening; however are easily avoidable with the proper knowledge and awareness. With our application, we aim to educate patients about the medication they are taking by highlighting important information, including side effects, allergy information, and any potential drug interactions. By increasing health literacy, we will support patient autonomy and hopefully decrease emergency room visits.

1.3 Solution

Our solution is a Medication Safety Assistant that helps users identify potential risks in their medications. With the input received from the patient/user, such as current medication, known allergies, and medication schedule, the application will analyze the medication list for potential drug interactions, allergy conflicts, and timing conflicts, such as medications that shouldn't be taken together or should be spaced apart. The application is not replacing professional medical advice, but is instead designed to improve the patient's awareness and understanding by providing digestible warnings and explanations about the possible medication safety concerns. Improving the health literacy of the patient will provide more informed discussions with their pharmacists and healthcare providers while also supporting safer medication use.

2 TECHNICAL DESIGN

2.1 Tools/Technology

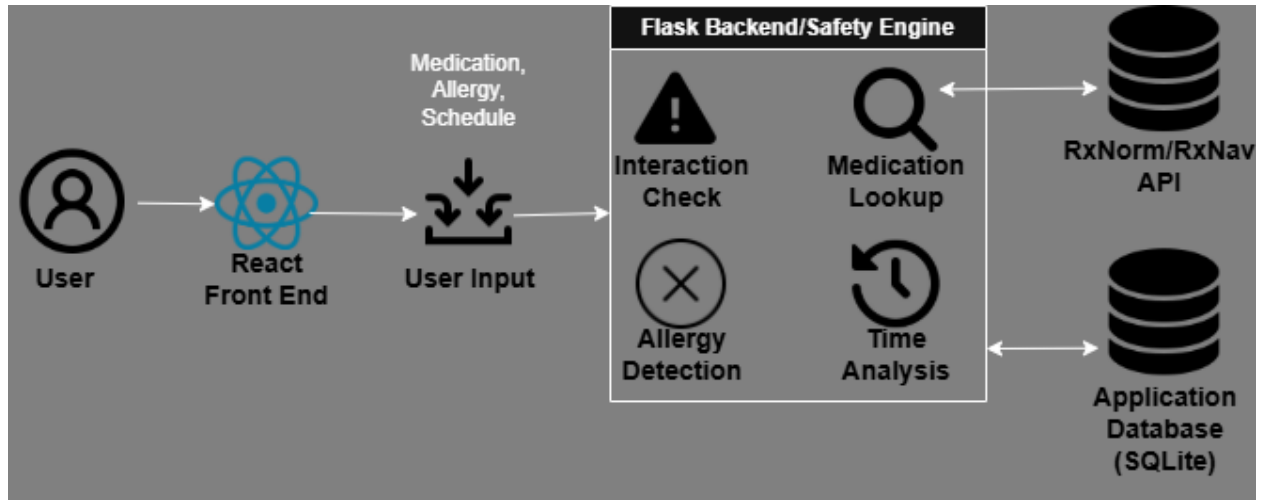
- **Python**
- **Flask**
- **React**
- **JavaScript**
- **HTML/CSS**
- **SQLite**
- **GitHub**
- **Figma**
- **RxNorm/RxNav API**

Python, Flask, React, Javascript, HTML/CSS, SQLite, Github, Figma, RxNorm/RxNav API

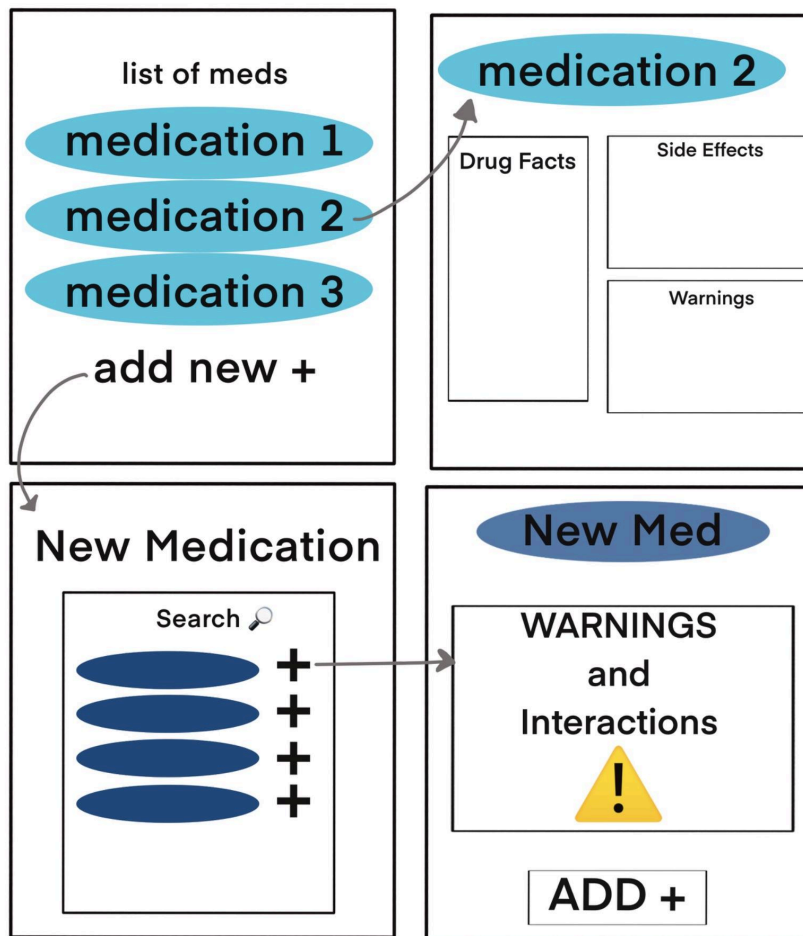
2.2 Datasets and Data Sources

Our application will use RxNorm / RxNav API as the data source for medication name lookup and normalization, ensuring medications entered by users are matched to the drug names within the application. We plan to create a curated rule set for interaction, allergy, and timing conflict checking. For testing, we will create synthetic patient scenarios with medication lists, allergies, and schedules to verify that the program correctly flags relevant warnings.

2.3 Architecture Diagram



2.4 Screen Mock-up(s)



3 IMPLEMENTATION PLAN

3.1 Project Tasks

3.2 Project Timeline

3.2 Needs/Risks

Task	Description	Needs/Risks	Timeline (week of semester)
Research Medication Data Sources RxNorm / RxNac API	We need to make sure we have access to medication information	Need: Reliable public medication data Risk: May not have full interaction data, requiring manual rule creation	Week 1
Design UI Wireframes	Create Figma wireframes for medication entry, profile setup, and warning displays	Risk: UI may need redesigns if requirements change	Weeks 1-2
Set Up Application Architecture	Create project repositories for frontend, backend, and database	Need: Team agreement on stack and development workflow Risk: Setup issues with React or backend/frontend integration	Week 2-3
Develop a User Profile	The app needs to be able to hold the user's medication information.	Need: Deciding what data to store and what to display. Avoid information dumping, but need to ensure important information is still being given to the user	Weeks 2-3

Develop a Medication Profile	This will be a combination of user input and editable information, as well as medication information that is NOT editable. Need to include basic drug information (how much, medication scheduling, ect.)	Risk: Input requirements may be more complex than expected	Weeks 3-4
Integrate Medication Lookup API	Connect backend to RxNorm/RxNav for medication lookup	Need: Documentation and working API access Risk: API limitations	Weeks 4-5
Implement Medication Safety Rules	Implement backend logic for interaction checks and allergy conflicts	Risk: Curated medication rules could take longer to create and validate than expected	Weeks 5-6
Implement Time Conflict Analysis	Implement logic for medications that should be spaced apart	Need: Reliable timing rules for the selected medications Risk: Limited public information available for some medications	Week 6
Implement 'add medication'	The user needs to be able to add to and edit their list of medications.	Risk: Can not be implemented before gaining access to the medication database	Week 6
Implement Warning system for drug interactions	Create a frontend display for interaction alerts, allergy warnings, and timing conflicts	Need: Make sure to reiterate that the app is NOT for medical advice, but still try to offer helpful warnings. Risk: This is one of the last tasks because it needs the users'	Weeks 6-7

		stored data and access to medication databases.	
Create Synthetic Test Cases	Create handcrafted patient scenarios with medication lists, allergies, and schedules for testing	Need: Representative test cases Risk: test cases may not cover enough edge cases	Week 7
Integration Testing and Bug Fixing	Test the full workflow from user input to backend analysis and display warnings	Risk: bugs could appear late in development	Week 7-8
Prepare Final Delivery	Prepare technical documentation, final report, and recorded presentation,	Risk: delays in previous tasks could result in less time to spend polishing	Week 8

4 References

Shehab N, Lovegrove MC, Geller AI, Rose KO, Weidle NJ, Budnitz DS. US Emergency Department Visits for Outpatient Adverse Drug Events, 2013-2014. *JAMA*. 2016;316(20):2115. doi:<https://doi.org/10.1001/jama.2016.16201>